Exploratory Data Analysis

Diabetes 130-US Hospitals for Years 1999–2008

1. Introduction

Exploratory Data Analysis (EDA) is an important first step in a machine learning workflow. Before selecting a machine learning algorithm or building a predictive model, it is necessary to understand the structure, quality, distribution, and limitations of the available data.

This project performs an exploratory analysis of the **Diabetes 130-US Hospitals for Years 1999–2008** dataset. The dataset contains information about diabetic patient hospital encounters collected from 130 US hospitals and integrated delivery networks over a ten-year period.

The original dataset is specifically concerned with early hospital readmission of diabetic patients. The UCI Machine Learning Repository identifies classification and clustering as associated machine learning tasks for this dataset.

The objectives of this analysis are:

- Understand the structure and characteristics of the dataset.
- Analyze the data types of different variables.
- Identify missing values and duplicate records.
- Calculate descriptive statistics.
- Analyze numerical and categorical variables.
- Detect potential statistical outliers.
- Examine relationships between numerical variables.
- Analyze the target variable and class distribution.
- Identify suitable machine learning problem formulations.
- Discuss limitations and ethical considerations associated with healthcare data.

2. Dataset Description

2.1 Dataset Name

Diabetes 130-US Hospitals for Years 1999–2008

2.2 Dataset Source

The dataset used in this project was obtained from the **Kaggle** dataset titled "Diabetes 130 US hospitals for years 1999–2008." The Kaggle dataset identifies the UCI Machine Learning Repository as the original source of the data.

The dataset represents ten years of clinical care from 1999 to 2008 at 130 US hospitals and integrated delivery networks.

- **Kaggle Dataset:**
  https://www.kaggle.com/datasets/brandao/diabetes
- **Original UCI Dataset:**
  UCI Machine Learning Repository
- **Dataset DOI:**
  10.24432/C5230J

The UCI repository identifies the subject area as Health and Medicine, with classification and clustering as associated machine learning tasks. The repository reports 101,766 instances and 47 features in its formal dataset description.

The downloaded CSV file used for this project contains **101,766 rows and 50 columns**. The difference between the UCI feature count and the CSV column count is due to the representation of identifier and target-related columns in the downloaded file.

3. Dataset Structure

The primary data file used for the analysis is:

`diabetic_data.csv`

The dataset contains information about diabetic patient hospital encounters, including demographic characteristics, admission information, diagnoses, procedures, medications, laboratory tests, healthcare utilization, and readmission outcomes.

The downloaded CSV contains:

- **101,766 rows**
- **50 columns**

Examples of variables include:

- `race`
- `gender`
- `age`
- `admission_type_id`
- `discharge_disposition_id`
- `admission_source_id`
- `time_in_hospital`

- num_lab_procedures
- num_procedures
- num_medications
- number_outpatient
- number_emergency
- number_inpatient
- number_diagnoses
- readmitted

4. Data Type Analysis

The dataset contains a combination of numerical and categorical variables.

4.1 Numerical Features

Numerical variables represent measurable quantities associated with a patient's hospital encounter.

Examples include:

- Time spent in hospital
- Number of laboratory procedures
- Number of medical procedures
- Number of medications
- Number of outpatient visits
- Number of emergency visits
- Number of inpatient visits
- Number of diagnoses

These variables can be analyzed using numerical summaries such as mean, median, standard deviation, minimum, maximum, and quartiles.

4.2 Categorical Features

Categorical variables represent groups or categories.

Examples include:

- Race
- Gender
- Age category
- Admission type

- Discharge disposition
- Admission source
- Medical specialty
- Diabetes medications
- Readmission status

Categorical variables can be analyzed using frequency counts, proportions, and bar charts.

5. Missing Value Analysis

Missing values are an important data-quality issue in healthcare datasets. A missing value does not necessarily mean that the information was incorrect. It may indicate that a measurement was not recorded or was not applicable to a particular patient encounter.

The analysis identified substantial missingness in several variables.

The most significant examples are:

- **`weight`: approximately 96.8% missing**
- **`medical_specialty`: approximately 49% missing**

The very high percentage of missing values in the `weight` variable means that this feature requires careful consideration before being used in a future machine learning model.

Similarly, the missing values in `medical_specialty` represent a substantial portion of the dataset.

5.1 Possible Causes of Missingness

Possible reasons for missing healthcare information include:

- A measurement was not required during a particular encounter.
- The information was not recorded.
- The information was unavailable.
- The variable was not applicable to a particular patient.
- Differences in hospital data-recording practices.

Missing values should therefore be investigated before automatically deleting rows or imputing values.

6. Duplicate Analysis

Duplicate records were examined during the exploratory analysis.

Duplicate rows can occur due to repeated data entry. However, repeated records involving the same patient do not necessarily represent duplicate observations because the dataset contains hospital encounters rather than only one record per patient.

Therefore, repeated patient identifiers should not automatically be treated as duplicate observations.

7. Descriptive Statistics

Descriptive statistics were calculated for numerical variables to understand their central tendency and spread.

The following statistical measures were considered:

- Count
- Mean
- Median
- Standard deviation
- Minimum
- Maximum
- First quartile (Q1)
- Second quartile (Q2)
- Third quartile (Q3)

These statistics help identify typical values and variability within the healthcare data.

Variables such as `time_in_hospital` and `num_medications` provide information about the duration and intensity of hospital encounters.

8. Numerical Feature Analysis

Numerical variables were examined using statistical summaries and distribution plots.

Important numerical variables include:

- `time_in_hospital`
- `num_lab_procedures`
- `num_procedures`
- `num_medications`
- `number_outpatient`
- `number_emergency`
- `number_inpatient`
- `number_diagnoses`

These variables describe healthcare utilization and characteristics of each hospital encounter.

The distributions of these variables are not necessarily symmetric. Several variables contain observations concentrated around typical ranges while also containing values at the extreme ends.

Such distributions are important when considering future preprocessing and machine learning model selection.

9. Categorical Feature Analysis

Categorical variables were analyzed using frequency distributions.

Examples include demographic characteristics, admission information, and medical treatment-related variables.

The categorical analysis helps identify:

- The most common categories.
- Rare categories.
- Imbalanced categorical variables.
- Potentially useful categorical predictors.

Categorical distributions should be examined carefully before encoding variables for machine learning because some categories may have very few observations.

10. Outlier Analysis

Outliers were identified using the **Interquartile Range (IQR)** method.

The IQR is calculated as:

$$IQR = Q_3 - Q_1$$

Potential lower and upper outlier boundaries are:

$$Lower\ Bound = Q_1 - 1.5(IQR)$$

$$Upper\ Bound = Q_3 + 1.5(IQR)$$

Observations outside these boundaries were treated as potential statistical outliers.

Significant numbers of potential outliers were identified in variables such as:

- `num_medications`
- `time_in_hospital`

10.1 Healthcare Interpretation of Outliers

In healthcare data, an extreme value may represent a genuine patient condition rather than an error.

For example, a patient requiring a large number of medications or a relatively long hospital stay may represent a genuine complex medical case.

Therefore, an outlier could be:

- A genuine severe medical case.
- A patient with multiple health conditions.
- A legitimate unusually long hospital stay.

- A data-entry error.
- A measurement or recording issue.

Consequently, outlier removal should only be performed after understanding the reason behind the extreme value.

11. Correlation Analysis

Correlation analysis was performed on numerical variables to identify potential relationships between features.

A correlation matrix and heatmap can be used to visualize these relationships.

Correlation analysis is useful because highly related variables may provide similar information. It can also help identify variables that may be useful for future predictive modeling.

However, correlation does not imply causation. A strong correlation between two variables does not necessarily mean that one variable causes the other.

12. Target Variable Analysis

The primary target variable considered for predictive analysis is `readmitted`.

The variable contains three readmission categories:

- `<30` – Readmitted within 30 days.
- `>30` – Readmitted after 30 days.
- `NO` – Not readmitted.

For a binary classification problem, the target can be reformulated to focus specifically on whether the patient was readmitted within 30 days.

The analysis found that approximately:

11.16% of encounters correspond to readmission within 30 days.

This indicates that the target is imbalanced when the task is defined as predicting 30-day readmission versus non-readmission.

12.1 Importance of Class Imbalance

Class imbalance is important because a model could achieve apparently high accuracy by mostly predicting the majority class.

For example, if only a small proportion of patients experience the target event, simply predicting "No" for nearly every patient could produce a misleadingly high accuracy while failing to identify patients who are actually readmitted.

Therefore, future classification work should consider appropriate evaluation metrics such as precision, recall, F1-score, and the confusion matrix rather than relying only on accuracy.

13. Possible Machine Learning Problem Types

The dataset can support multiple machine learning problem formulations. The appropriate problem depends on the target selected from the available variables.

13.1 Classification

Classification is strongly supported by the dataset.

The original UCI repository identifies classification as one of the associated tasks for this dataset.

A possible classification problem is:

Predict whether a patient will be readmitted within 30 days.

The target can be represented as:

- **1** – Readmitted within 30 days.
- **0** – Not readmitted within 30 days.

Possible features include:

- Age
- Time spent in hospital
- Number of previous inpatient visits
- Number of emergency visits
- Number of medications
- Number of diagnoses
- Number of laboratory procedures
- Admission information

Classification is appropriate because the target represents a categorical outcome.

13.2 Regression

Regression is **not one of the officially listed associated tasks** for the original UCI dataset. However, the available variables can be used to formulate a separate regression problem.

For example:

Predict the length of a patient's hospital stay.

The variable `time_in_hospital` provides a numerical target that can be predicted using other patient and encounter characteristics.

Possible input features include:

- Age
- Number of diagnoses

- Number of procedures
- Number of laboratory procedures
- Number of medications
- Admission type
- Previous hospital utilization

Regression is appropriate when the objective is changed from predicting a categorical readmission outcome to predicting a numerical quantity.

Therefore, regression is considered a **possible reformulation** of the dataset rather than an officially associated task.

13.3 Clustering

Clustering is also supported by the dataset.

The UCI repository identifies clustering as an associated machine learning task.

A clustering problem could group patients or hospital encounters according to similarities in healthcare utilization and patient characteristics.

Possible clustering features include:

- Age
- Number of medications
- Number of diagnoses
- Number of inpatient visits
- Number of emergency visits
- Number of outpatient visits
- Length of hospital stay

Unlike classification, clustering does not require a predefined target label. The objective would be to discover natural groups within the dataset.

For example, clustering could potentially identify groups of patients with different levels of healthcare utilization.

14. Summary of Suitable ML Problems

ML Problem	**Status**	**Example**
Classification	Strongly Supported	Predict 30-day hospital readmission
Regression	Possible Reformulation	Predict length of hospital stay
Clustering	Supported	Group patients based on healthcare utilization

15. Key EDA Findings

The major findings from the exploratory analysis are:

1. The dataset contains **101,766 records and 50 columns** in the downloaded CSV file.
2. The dataset contains a combination of numerical and categorical variables, allowing multiple forms of exploratory analysis.
3. Missing values are a significant data-quality issue. The `weight` feature has approximately **96.8% missing values**, while `medical_specialty` has approximately **49% missing values**.
4. Statistical outliers were identified in variables such as `num_medications` and `time_in_hospital`.
5. The identified outliers should not automatically be removed because extreme values may represent genuine complex healthcare cases.
6. The `readmitted` variable provides a meaningful target for a classification problem.
7. Approximately **11.16%** of the records correspond to readmission within 30 days, indicating class imbalance for a binary 30-day readmission task.
8. The dataset officially supports classification and clustering, while regression can be formulated using a suitable numerical target.

16. Limitations and Ethical Considerations

Healthcare data requires additional consideration because incorrect analysis or predictions can potentially affect patients and healthcare decisions.

16.1 Missing and Incomplete Data

The high percentage of missing values in some variables can introduce uncertainty or reduce the usefulness of those features. Missingness should be understood before choosing an imputation or removal strategy.

16.2 Dataset Bias

The dataset represents hospital encounters from a particular period and healthcare setting. Therefore, its patterns may not necessarily represent all hospitals, countries, populations, or modern healthcare environments.

16.3 Sensitive Information

The UCI repository notes that the dataset contains information that may be considered sensitive, including patient age, gender, and race.

Any use of healthcare data should therefore follow appropriate privacy and ethical practices.

16.4 Fairness

A predictive model could perform differently across demographic groups if the training data is not sufficiently representative. This could lead to unfair or inaccurate predictions.

16.5 Human Oversight

Machine learning predictions should support healthcare professionals rather than replace clinical judgment. A model's prediction should not be treated as an unquestionable medical decision.

17. Conclusion

This exploratory data analysis examined the **Diabetes 130-US Hospitals for Years 1999–2008** dataset, which contains 101,766 hospital encounter records in the downloaded CSV file.

The analysis demonstrated that understanding data quality is essential before developing a machine learning model. The dataset contains both numerical and categorical features, substantial missing values, and statistical outliers.

In particular, the `weight` and `medical_specialty` variables contain significant missing data. Potential outliers were also identified in variables such as `num_medications` and `time_in_hospital`.

The `readmitted` variable provides a suitable basis for a classification problem focused on predicting 30-day readmission. The dataset also supports clustering according to the original UCI description. A regression problem can additionally be formulated using a numerical variable such as hospital length of stay, although regression is not an officially listed associated task for the original dataset.

Overall, this EDA demonstrates that machine learning should begin with understanding the data rather than immediately selecting an algorithm. Analyzing structure, quality, distributions, missing values, outliers, and potential targets provides the foundation for developing reliable and responsible machine learning solutions.

18. Final Dataset Profile

Aspect	Finding
Dataset	Diabetes 130-US Hospitals for Years 1999–2008
Dataset Source	Kaggle / UCI Machine Learning Repository
Records	101,766
CSV Columns	50
UCI Reported Features	47
Data Types	Numerical and categorical variables
Major Missing Values	Weight: approximately 96.8%; Medical Specialty: approximately 49%
Potential Outliers	Identified in variables including num_medications and time_in_hospital
Primary Target	`readmitted`
30-Day Readmission	Approximately 11.16% of records
Classification	Supported – predict 30-day readmission
Regression	Possible reformulation – predict length of hospital stay
Clustering	Supported – group patients based on healthcare utilization

19. References

1. Clore, J., Cios, K., DeShazo, J., & Strack, B. (2014). *Diabetes 130-US Hospitals for Years 1999–2008*. UCI Machine Learning Repository.

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4. Strack, B., DeShazo, J. P., Gennings, C., Olmo, J. L., Ventura, S., Cios, K. J., & Clore, J. N. (2014). Impact of HbA1c Measurement on Hospital Readmission Rates: Analysis of 70,000 Clinical Database Patient Records. *BioMed Research International*, 2014.

20. Dataset Links

- **Kaggle:** Diabetes 130 US Hospitals for Years 1999–2008
- **UCI Machine Learning Repository:** Diabetes 130-US Hospitals for Years 1999–2008
- **Dataset DOI:** https://doi.org/10.24432/C5230J

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